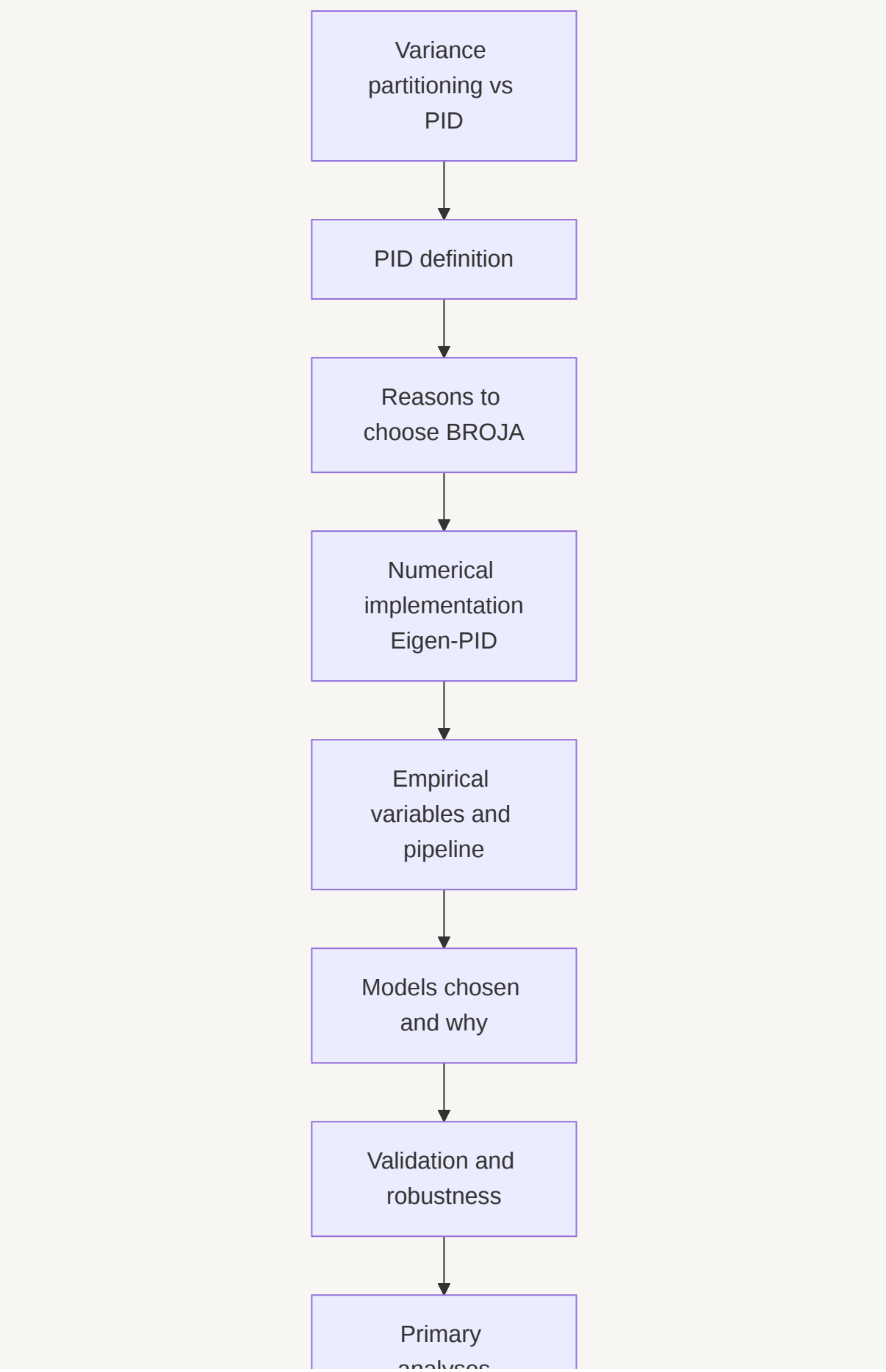


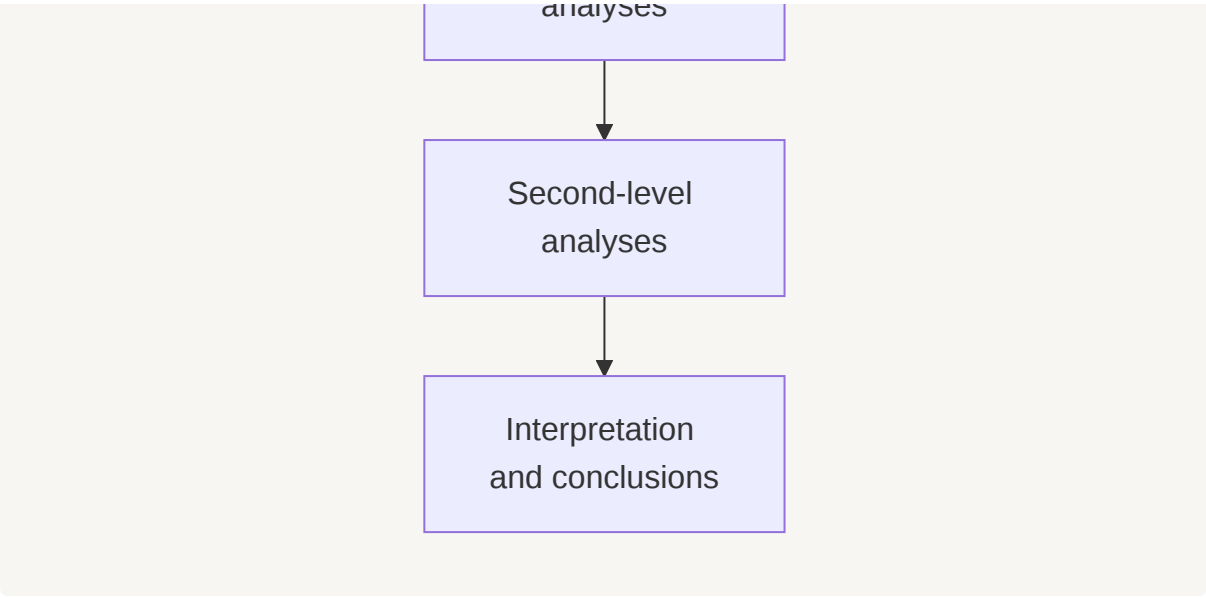
# PAPER FLOW — DNN–Brain PID



This is a living guide to the paper's scientific argument and a checklist for planning the work. It is not manuscript text. Keep the order here synchronized with the linked pages whenever the story changes.

## Paper flow


Variance partitioning vs PID
PID definition
Reasons to choose BROJA
Numerical implementation Eigen-PID
Empirical variables and pipeline
Models chosen and why
Validation and robustness
Primary




## How to maintain this notebook

1. Update the flowchart before changing the planned paper order.
2. Add, remove, or rename the corresponding linked page.
3. Update that page's current knowledge, missing decisions, and checklist.
4. Keep detailed mathematics, code, and experiment logs in their existing pages and link to them here.

## Current state

| Stage | Current state |
|---|---|
| VP vs PID | Rationale established; presentation still to design |
| PID and BROJA | Framework selected; concise paper argument still to finalize |
| Eigen-PID | Theoretical contribution completed; role in this paper to shape |
| Empirical pipeline | Planned and partly implemented; methodological choices remain |
| Model selection | In progress |
| Primary and second-level analyses | Planned; final analyses and statistics remain open |

## Notebook pages

01 — Variance Partitioning vs PID

02 — PID Definition

03 — Reasons to Choose BROJA

04 — Numerical Implementation: Eigen-PID

05 — Empirical Variables and Pipeline

06 — Models Chosen and Why

07 — Validation and Robustness

08 — Primary Analyses

09 — Second-Level Analyses

10 — Interpretation and Conclusions